8. 3

1

(3) ∨ $\int (x^3+1)\ln x\ dx$

$$\left[\ln x\cdot\left(\frac{1}{4}x^4+x\right)\right]-\int\left(\frac{1}{4}x^3+1\right)dx$$

$$=\left(\frac{1}{4}x^4+x\right)\ln x-\frac{1}{16}x^4-x+C$$

(8) ∨ $\int \ln^3 x\ dx$

$$=\left[\ln^3 x\cdot x\right]-\int 3\ln^2 x\cdot\frac{1}{x}\cdot x\ dx$$

$$=x\ln^3 x-3\int\ln^2 x\ dx$$

$$=x\ln^3 x-3\left(\left[\ln^2 x\cdot x\right]-\int 2\ln x\ dx\right)$$

$$=x\ln^3 x-3x\ln^2 x+6(x\ln x-x)+C$$

$$=x\ln^3 x-3x\ln^2 x+6x\ln x-6x+C\ //$$

(13) ∨ $\int x\sin^{-1}x\ dx$

$$=\sin^{-1}x\cdot\frac{1}{2}x^2-\int\frac{1}{2}x^2\cdot\frac{1}{\sqrt{1-x^2}}\ dx$$

$u=\sqrt{1-x^2}$ $\quad u^2=1-x^2$ $\quad x^2=1-u^2$

$$\frac{-2x}{2\sqrt{1-x^2}}\ dx=du$$

$$\sin^{-1}x\cdot\frac{1}{2}x^2+\frac{1}{2}\int\sqrt{1-u^2}\ du$$

$$=\sin^{-1}x\cdot\frac{1}{2}x^2+\frac{1}{2}\cdot\frac{2}{3}(1-u^2)^{\frac{3}{2}}\cdot\left(-\frac{1}{2u}\right)$$

$$=\sin^{-1}x\cdot\frac{1}{2}x^2+\frac{1}{3}x^3\cdot\left(-\frac{1}{2\sqrt{1-x^2}}\right)$$

$$=\sin^{-1}x\cdot\frac{1}{2}x^2-\frac{1}{6}\cdot\frac{x^3}{\sqrt{1-x^2}}\ //$$

2

(2) ∨ $\int_0^1 x^3\sin(x^2)\ dx$

$x^2=t$

$2x\,dx=dt$

$$\frac{1}{2}\int_0^1 t\sin t\ dt=\frac{1}{2}\left[t\cdot(-\cos t)-(-\sin t)\right]_0^1$$

$$=\frac{1}{2}(\sin 1-\cos 1)\ //$$

2

(6) ∨ $\int_1^2 (x^3+3)\ln x\,dx$

$= \left[\ln x\cdot\left(\frac{1}{4}x^4+3x\right)\right]_1^2 - \int_1^2 \frac{1}{4}x^3+3\,dx$

$= 10\cdot\ln 2 - \left[\frac{1}{16}x^4+3x\right]_1^2$

$= 10\ln 2 - \left(1+6-\frac{1}{16}-3\right)$

$= 10\ln 2 - \frac{63}{16}$ //

3 ∨

(3) $\int \sec^n x\,dx$

$= \int \sec^{n-2}x\cdot\sec^2 x\,dx$

$= \sec^{n-2}x\cdot\tan x - \int (n-2)\sec^{n-2}x\cdot\tan^2 x\,dx$

$= -\int (n-2)\sec^{n-2}x\cdot(\sec^2 x-1)\,dx$

$= -\int (n-2)\sec^{n-2}x\cdot\sec^2 x + \int (n-2)\sec^{n-2}x\,dx$

$\int (n-2+1)\sec^{n-2}x\cdot\sec^2 x\,dx$

$= \sec^{n-2}x\tan x + \int (n-2)\sec^{n-2}x\,dx$

$\therefore \int \sec^{n-2}x\sec^2 x\,dx = \frac{1}{n-1}\sec^{n-2}\tan x + \frac{n-2}{n-1}\int \sec^{n-2}x\,dx$

$= \int \sec^n x\,dx$